

Practical-02: --

A)

Code Output: -

```
aswitha_patel@Aswitha:~$ nano prac02.c
aswitha_patel@Aswitha:~$ gcc prac02.c -o prac02
aswitha_patel@Aswitha:~$ ./prac02
Files successfully opened
File copied successfully
aswitha_patel@Aswitha:~$ |
```

Contents of source file

```
aswitha_patel@Aswitha:~$ cat source.txt
this is aswitha
Studying in K LH University
aswitha_patel@Aswitha:~$ |
```

Contents of destination file

```
aswitha_patel@Aswitha:~$ cat destination.txt
this is aswitha
Studying in K LH University
aswitha_patel@Aswitha:~$ |
```

B)

Command

```
aswitha_patel@Aswitha:~$ strace cat sample.txt
execve("/usr/bin/cat", ["cat", "sample.txt"], 0x7ffd262442c8 /* 25 vars */) = 0
brk(NULL) = 0x56f6c8aa0000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7995022fc000
access("/etc/ld.so.preload", R_OK) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=20085, ...}) = 0
mmap(NULL, 20085, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7995022f7000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=199120, ...}) = 0
mmap(NULL, 206288, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7995022c4000
mmap(0x7995022c0000, 135168, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x7000) = 0x7995022cb000
mmap(0x7995022ec000, 28672, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x7995022ec000
mmap(0x7995022f3000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2f000) = 0x7995022f3000
mmap(0x7995022f5000, 5584, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x7995022f5000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libgcc_s.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=187120, ...}) = 0
mmap(NULL, 185256, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x799502296000
mmap(0x79950229a000, 147456, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x4000) = 0x79950229a000
mmap(0x7995022be000, 16384, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x7995022be000
mmap(0x7995022c2000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2c000) = 0x7995022c2000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libm.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=1198376, ...}) = 0
mmap(NULL, 1200152, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x799502170000
```

```
mmap(0x799502182000, 655360, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x12000) = 0x799502182000
mmap(0x799502222000, 466944, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0xb2000) = 0x799502222000
mmap(0x799502294000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x123000) = 0x799502294000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0 \250\2\0\0\0\0"..., 832) = 832
pread64(3, "\6\0\0\0\4\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0"..., 840, 64) = 840
fstat(3, {st_mode=S_IFREG|0755, st_size=2186512, ...}) = 0
pread64(3, "\6\0\0\0\4\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0"..., 840, 64) = 840
mmap(NULL, 2231696, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x799501e00000
mmap(0x799501e28000, 1671168, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x799501e28000
mmap(0x799501fc0000, 319488, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1c0000) = 0x799501fc0000
mmap(0x79950200e000, 24576, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x20d000) = 0x79950200e000
mmap(0x799502014000, 52624, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x799502014000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libpcre2-8.so.0", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=699056, ...}) = 0
mmap(NULL, 701112, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7995020c4000
mmap(0x7995020c7000, 507904, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x3000) = 0x7995020c7000
mmap(0x799502143000, 176128, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x7f000) = 0x799502143000
mmap(0x79950216e000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0xa9000) = 0x79950216e000
close(3) = 0
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7995020c2000
mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7995020bfb000
arch_prctl(ARCH_SET_FS, 0x7995020cbfb00) = 0
set_tid_address(0x7995020c0128) = 781
set_robust_list(0x7995020bfbde0, 24) = 0
rseq(0x7995020bfb6a0, 0x20, 0, 0x53053053) = 0
mprotect(0x79950200e000, 16384, PROT_READ) = 0
```

```
mprotect(0x7995022c2000, 4096, PROT_READ) = 0
mprotect(0x7995022f3000, 4096, PROT_READ) = 0
mprotect(0x56f6beb36000, 1462272, PROT_READ) = 0
mprotect(0x799502341000, 8192, PROT_READ) = 0
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINITY}) = 0
getrandom("\x16\xfe\x10\xbe\xec\xfe\x8\x5e", 8, GRND_NONBLOCK) = 8
munmap(0x7995022f7000, 20085) = 0
statfs("/sys/fs/selinux", {f_type=SYSFS_MAGIC, f_bsize=4096, f_blocks=0, f_bfree=0, f_bavail=0, f_files=0, f_ffree=0, f_fsid={val=[0x2fdf4c0d, 0xb5ad53f4]}, f_namelen=255, f_frsize=4096, f_flags=ST_NOSUID|ST_NODEV|ST_NOEXEC|ST_NOATIME}) = 0
statfs("/selinux", 0x7ffd3b13c340) = -1 ENOENT (No such file or directory)
brk(NULL) = 0x56f6c8aa0000
brk(0x56f6c8ac1000) = 0x56f6c8ac1000
openat(AT_FDCWD, "/proc/filesystems", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0444, st_size=0, ...}) = 0
read(3, "nodev\tsysfs\nnodev\ttmpfs\nnodev\tbdf"..., 1024) = 442
lseek(3, 0, SEEK_CUR) = 442
lseek(3, 416, SEEK_SET) = 416
close(3) = 0
openat(AT_FDCWD, "/proc/mounts", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0444, st_size=0, ...}) = 0
read(3, "none /usr/lib/modules/6.18.33.2-"..., 1024) = 1024
read(3, "e 0 0\ndevpts /dev/pts devpts rw,"..., 1024) = 1024
read(3, "osuid,nodev,noexec,relatime,nosy"..., 1024) = 1024
read(3, "s /mnt/wslg/run/user/1000 tmpfs "..., 1024) = 118
read(3, "", 1024) = 0
close(3) = 0
access("/etc/selinux/config", F_OK) = -1 ENOENT (No such file or directory)
fcntl(0, F_GETFD) = 0
fcntl(1, F_GETFD) = 0
```

```

fcntl(1, F_GETFD) = 0
fcntl(2, F_GETFD) = 0
rt_sigaction(SIGPIPE, NULL, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 8) = 0
poll([{fd=0, events=0}, {fd=1, events=0}, {fd=2, events=0}], 3, 0) = 0 (Timeout)
rt_sigaction(SIGPIPE, {sa_handler=SIG_IGN, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x799501e45cb0}, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 8) = 0
openat(AT_FDCWD, "/proc/self/maps", O_RDONLY|O_CLOEXEC) = 3
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIMIT64_INFINITY}) = 0
fstat(3, {st_mode=S_IFREG|0444, st_size=0, ...}) = 0
read(3, "56f6be1cd000-56f6be2bf000 r--p 0"... , 1024) = 1024
read(3, "0 08:30 14176" , 1024) = 1024
read(3, "00 08:30 14388" , 1024) = 1024
read(3, " /usr/lib/x86_64-linux"... , 1024) = 1024
read(3, "0 13931 /usr"... , 1024) = 566
close(3) = 0
sched_getaffinity(781, 32, [0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15]) = 8
rt_sigaction(SIGSEGV, NULL, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 8) = 0
sigaltstack(NULL, {ss_sp=NULL, ss_flags=SS_DISABLE, ss_size=0}) = 0
mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK, -1, 0) = 0x7995022f9000
mprotect(0x7995022f9000, 4096, PROT_NONE) = 0
sigaltstack({ss_sp=0x7995022fa000, ss_flags=0, ss_size=8192}, NULL) = 0
gettid() = 781
rt_sigaction(SIGSEGV, {sa_handler=0x56f6be4f20c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x799501e45cb0}, NULL, 8) = 0
rt_sigaction(SIGBUS, NULL, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 8) = 0
rt_sigaction(SIGBUS, {sa_handler=0x56f6be4f20c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x799501e45cb0}, NULL, 8) = 0
prctl(PR_GET_AUXV, 0x7ffd3b13be30, 0x200, 0, 0) = 448
statx(AT_FDCWD, "/usr/share/coreutils/locales/cat", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFDIR|0755, stx_size=4096, ...}) = 0

```

```

0, stx_attributes=0, stx_mode=S_IFDIR|0755, stx_size=4096, ...}) = 0
openat(AT_FDCWD, "/usr/share/coreutils/locales/uucore/en-US.ftl", O_RDONLY|O_CLOEXEC) = 3
statx(3, "", AT_STATX_SYNC_AS_STAT|AT_EMPTY_PATH, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFREG|0644, stx_size=4080, ...}) = 0
read(3, "# Common strings shared across a"... , 4080) = 4080
read(3, "", 32) = 0
close(3) = 0
statx(AT_FDCWD, "/usr/share/coreutils/locales/cat", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFDIR|0755, stx_size=4096, ...}) = 0
openat(AT_FDCWD, "/usr/share/coreutils/locales/cat/en-US.ftl", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
rt_sigaction(SIGPIPE, {sa_handler=SIG_DFL, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x799501e45cb0}, {sa_handler=SIG_IGN, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x799501e45cb0}, 8) = 0
rt_sigaction(SIGSEGV, {sa_handler=SIG_DFL, sa_mask=~[RTMIN RT_1], sa_flags=SA_RESTORER, sa_restorer=0x799501e45cb0}, {sa_handler=0x56f6be4f20c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x799501e45cb0}, 8) = 0
rt_sigaction(SIGBUS, {sa_handler=SIG_DFL, sa_mask=~[RTMIN RT_1], sa_flags=SA_RESTORER, sa_restorer=0x799501e45cb0}, {sa_handler=0x56f6be4f20c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x799501e45cb0}, 8) = 0
ioctl(1, TCGETS2, {c_iflag=ICRNL|IXON, c_oflag=NL0|CR0|TAB0|BS0|VT0|FF0|OPOST|ONLCR, c_cflag=B38400|CS8|CREAD, c_lflag=ISIG|ICANON|ECHO|ECHOE|ECHOK|IEXTEN|ECHOCTL|ECHOKE, ...}) = 0
brk(0x56f6c8ae2000) = 0x56f6c8ae2000
statx(AT_FDCWD, "sample.txt", AT_STATX_SYNC_AS_STAT, STATX_ALL, 0x7ffd3b12b780) = -1 ENOENT (No such file or directory)
write(2, "cat", 3cat) = 3
write(2, ": ", 2:) = 2
write(2, "sample.txt: No such file or dire"... , 37sample.txt: No such file or directory) = 37
write(2, "\n", 1) = 1
sigaltstack({ss_sp=NULL, ss_flags=SS_DISABLE, ss_size=8192}, NULL) = 0
munmap(0x7995022f9000, 12288) = 0
exit_group(1) = ?
+++ exited with 1 +++

```

```

openat(AT_FDCWD, "/usr/share/coreutils/locales/uucore/en-US.ftl", O_RDONLY|O_CLOEXEC) = 3
statx(3, "", AT_STATX_SYNC_AS_STAT|AT_EMPTY_PATH, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFREG|0644, stx_size=4080, ...}) = 0
read(3, "# Common strings shared across a"... , 4080) = 4080
read(3, "", 32) = 0
close(3) = 0
statx(AT_FDCWD, "/usr/share/coreutils/locales/cat", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFDIR|0755, stx_size=4096, ...}) = 0
openat(AT_FDCWD, "/usr/share/coreutils/locales/cat/en-US.ftl", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
rt_sigaction(SIGPIPE, {sa_handler=SIG_DFL, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x799501e45cb0}, {sa_handler=SIG_IGN, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x799501e45cb0}, 8) = 0
rt_sigaction(SIGSEGV, {sa_handler=SIG_DFL, sa_mask=~[RTMIN RT_1], sa_flags=SA_RESTORER, sa_restorer=0x799501e45cb0}, {sa_handler=0x56f6be4f20c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x799501e45cb0}, 8) = 0
rt_sigaction(SIGBUS, {sa_handler=SIG_DFL, sa_mask=~[RTMIN RT_1], sa_flags=SA_RESTORER, sa_restorer=0x799501e45cb0}, {sa_handler=0x56f6be4f20c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x799501e45cb0}, 8) = 0
ioctl(1, TCGETS2, {c_iflag=ICRNL|IXON, c_oflag=NL0|CR0|TAB0|BS0|VT0|FF0|OPOST|ONLCR, c_cflag=B38400|CS8|CREAD, c_lflag=ISIG|ICANON|ECHO|ECHOE|ECHOK|IEXTEN|ECHOCTL|ECHOKE, ...}) = 0
brk(0x56f6c8ae2000) = 0x56f6c8ae2000
statx(AT_FDCWD, "sample.txt", AT_STATX_SYNC_AS_STAT, STATX_ALL, 0x7ffd3b12b780) = -1 ENOENT (No such file or directory)
write(2, "cat", 3cat) = 3
write(2, ": ", 2:) = 2
write(2, "sample.txt: No such file or dire"... , 37sample.txt: No such file or directory) = 37
write(2, "\n", 1) = 1
sigaltstack({ss_sp=NULL, ss_flags=SS_DISABLE, ss_size=8192}, NULL) = 0
munmap(0x7995022f9000, 12288) = 0
exit_group(1) = ?
+++ exited with 1 +++
aswitha_patel@Aswitha:~$ |

```